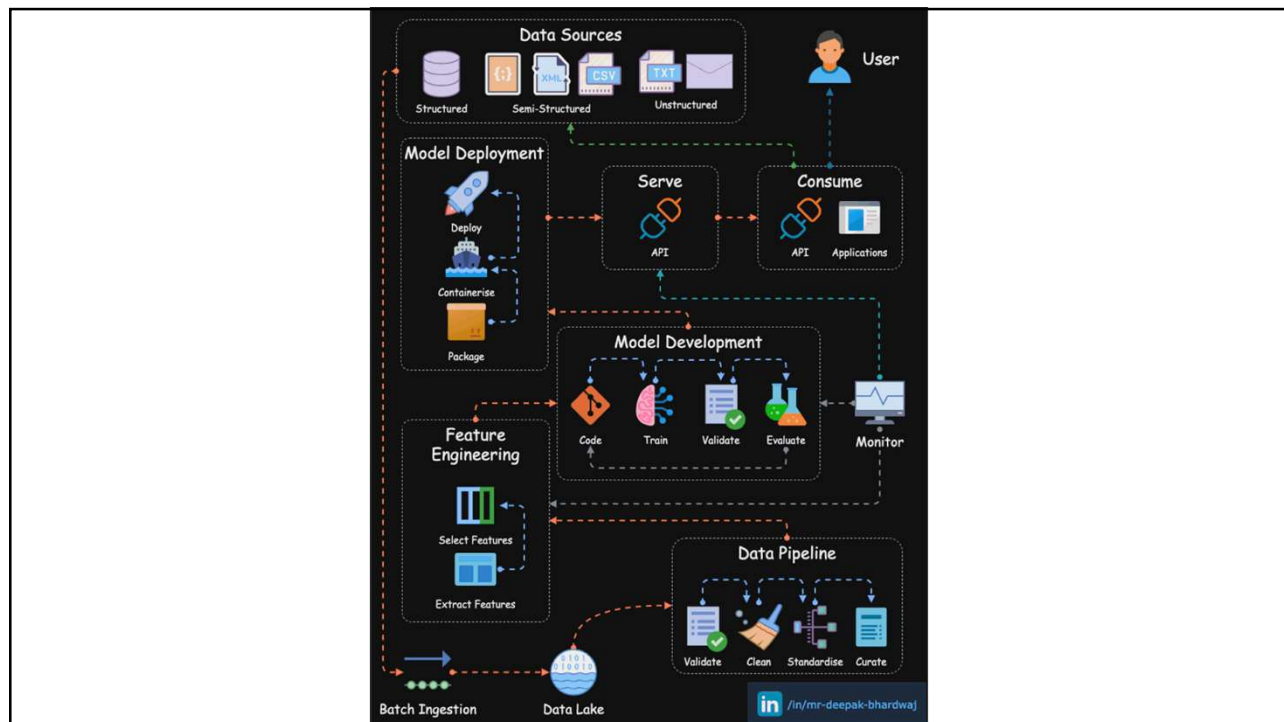






Agenda

- Tipos de datos
- Almacenes de datos
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CustomerID	Title	FirstName	MiddleName	LastName	Suffix	CompanyName	Phone
1	Mr.	Orlando	N.	Ge	NULL	A Bike Store	245-555-0173
2	Mr.	Keth	NULL	Harris	NULL	Progressive Sports	170-555-0127
3	Ms.	Donna	F.	Cameras	NULL	Advanced Bike Components	279-555-0130
4	Ms.	Jonet	M.	Gates	NULL	Modular Cycle Systems	710-555-0173
5	Mr.	Lucy	NULL	Harrington	NULL	Metropolitan Sports Supply	828-555-0186
6	Ms.	Rosmarie	J.	Carroll	NULL	Aerobic Exercise Company	244-555-0112
7	Mr.	Dominic	P.	Gash	NULL	Associated Bikes	152-555-0173
10	Ms.	Kathleen	M.	Garza	NULL	Rural Cycle Emporium	150-555-0127
11	Ms.	Katherine	NULL	Harding	NULL	Sharp Bikes	526-555-0159
12	Mr.	Johnny	A.	Caprio	Jr.	Bikes and Motorbikes	112-555-0191
16	Mr.	Christopher	R.	Beck	Jr.	Bulk Discount Store	1 (11) 500 555-0132
18	Mr.	David	J.	Lu	NULL	Catalog Store	440-555-0132
19	Mr.	John	A.	Beaver	NULL	Center Cycle Shop	521-555-0195
20	Ms.	Jean	P.	Handley	NULL	Central Discount Store	582-555-0113
21	N.	Jinghao	NULL	Lu	NULL	Chic Department Stores	528-555-0116
22	Ms.	Linda	E.	Burnett	NULL	Travel Systems	121-555-0121
23	Mr.	Kenn	NULL	Hanf	NULL	Bike World	216-555-0122
24	Mr.	Kevin	NULL	Lu	NULL	Eastside Department Store	526-555-0164
25	Mr.	Donald	L.	Stanton	NULL	Coalition Bike Company	367-555-0161
28	Ms.	Jackie	E.	Blackwell	NULL	Commuter Bicycle Store	972-555-0163
29	Mr.	Bryan	NULL	Hamilton	NULL	Cross-Country Riding Supp...	344-555-0144
30	Mr.	Todd	R.	Logan	NULL	Cycle Merchants	783-555-0110
34	Ms.	Barbara	J.	German	NULL	Cycles Wholesaler & Mfg.	1 (11) 500 555-0181
37	Mr.	Jim	NULL	Gest	NULL	Two Bike Shops	724-555-0161

Datos estructurados

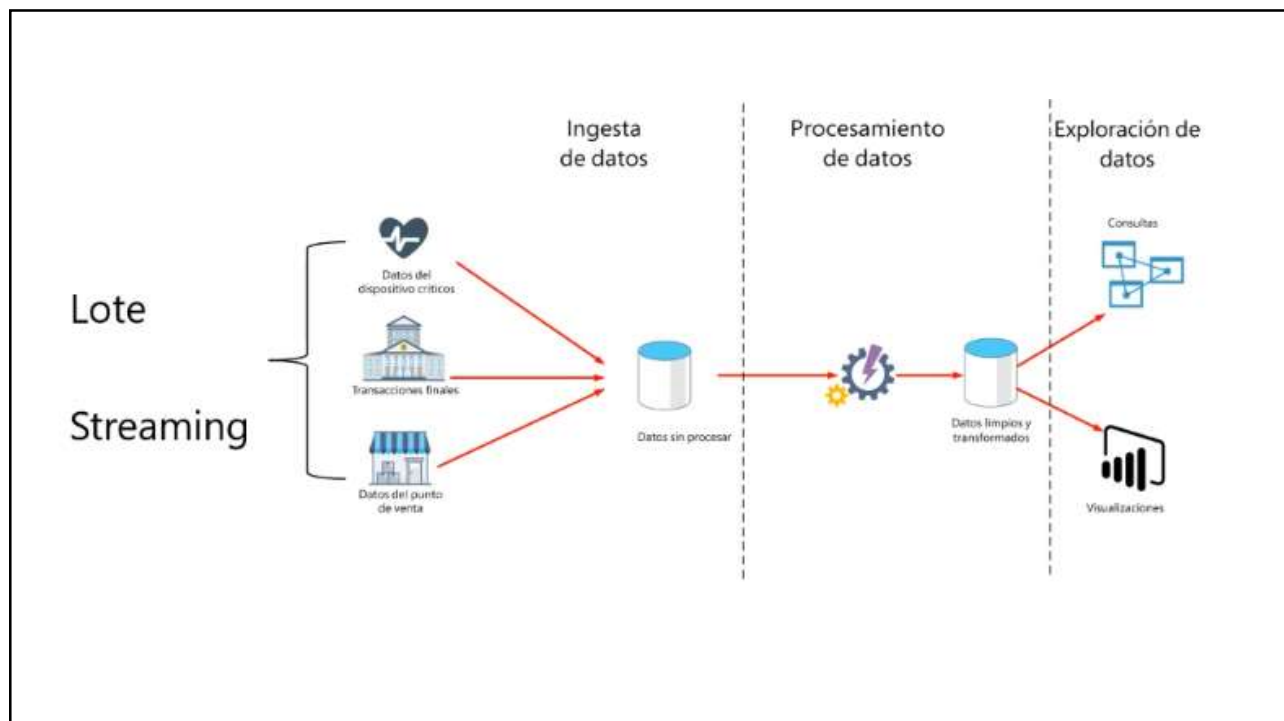
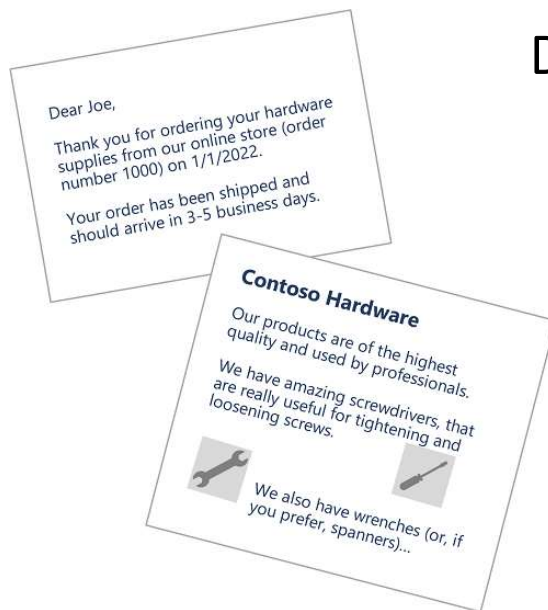
ProductID	Name	ProductNumber	Color	StandardCost	ListPrice	Size	Weight
680	HL Road Frame - Black, 58	FR-RS2B-58	Black	1059.31	1431.50	58	1016.04
706	HL Road Frame - Red, 58	FR-RS2R-58	Red	1059.31	1431.50	58	1016.04
707	Spot-100 Helmet, Red	HL-U509-R	Red	13.0863	34.99	NULL	NULL
708	Spot-100 Helmet, Black	HL-U509	Black	13.0863	34.99	NULL	NULL
709	Mountain Bike Socks, M	SO-B309-M	White	3.3963	9.50	M	NULL
710	Mountain Bike Socks, L	SO-B309-L	White	3.3963	9.50	L	NULL
711	Spot-100 Helmet, Blue	HL-U509-B	Blue	13.0863	34.99	NULL	NULL
712	AIRC Logo Cap	CA-1090	Multi	6.9223	8.99	NULL	NULL
713	Long-Sleeve Logo Jersey, S	LJ-0192-S	Multi	38.4923	49.99	S	NULL
714	Long-Sleeve Logo Jersey, M	LJ-0192-M	Multi	38.4923	49.99	M	NULL
715	Long-Sleeve Logo Jersey, L	LJ-0192-L	Multi	38.4923	49.99	L	NULL
716	Long-Sleeve Logo Jersey, XL	LJ-0192-X	Multi	38.4923	49.99	XL	NULL
717	HL Road Frame - Red, 62	FR-RS2R-62	Red	868.6342	1431.50	62	1043.26
718	HL Road Frame - Red, 44	FR-RS2R-44	Red	868.6342	1431.50	44	961.61
719	HL Road Frame - Red, 48	FR-RS2R-48	Red	868.6342	1431.50	48	979.75
720	HL Road Frame - Red, 52	FR-RS2R-52	Red	868.6342	1431.50	52	997.90
721	HL Road Frame - Red, 56	FR-RS2R-56	Red	868.6342	1431.50	56	1016.04
722	LL Road Frame - Black, 58	FR-R38B-58	Black	204.6251	337.22	58	1115.03
723	LL Road Frame - Black, 60	FR-R38B-60	Black	204.6251	337.22	60	1124.90
724	LL Road Frame - Black, 62	FR-R38B-62	Black	204.6251	337.22	62	1133.98
725	LL Road Frame - Red, 44	FR-R38R-44	Red	187.1571	337.22	44	1052.33
726	LL Road Frame - Red, 48	FR-R38R-48	Red	187.1571	337.22	48	1070.47

```
// Customer 1
{
  "firstName": "Joe",
  "lastName": "Jones",
  "address":
  {
    "streetAddress": "1 Main St.",
    "city": "New York",
    "state": "NY",
    "postalCode": "10099"
  },
  "contact":
  [
    {
      "type": "home",
      "number": "555 123-1234"
    },
    {
      "type": "email",
      "address": "joe@litware.com"
    }
  ]
}
```

```
// Customer 2
{
  "firstName": "Samir",
  "lastName": "Nadoy",
  "address":
  {
    "streetAddress": "123 Elm Pl.",
    "unit": "500",
    "city": "Seattle",
    "state": "WA",
    "postalCode": "98999"
  },
  "contact":
  [
    {
      "type": "email",
      "address": "samir@northwind.com"
    }
  ]
}
```

Datos semiestructurados

Datos no estructurados



Agenda

- Tipos de datos
- **Almacenes de datos**
 - **Almacenes de archivos**
 - Bases de datos
 - OLTP - OLAP
- Desarrollo de modelos

2. Almacenamiento de archivos

- El formato de archivo depende de :
 - El tipo de datos (estructurados, semiestructurados o no estructurados).
 - Las aplicaciones y los servicios que tendrán que leer, escribir y procesar los datos.
 - La necesidad de que los archivos de datos sean legibles para los usuarios o estén optimizados para un almacenamiento y procesamiento eficientes.

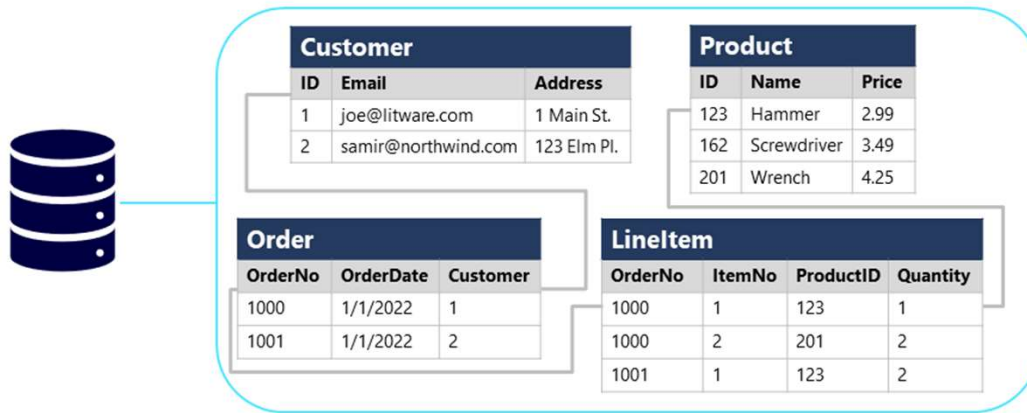
Formatos de archivo optimizados

- Los formatos legibles no están optimizados para el procesamiento ni almacenamiento.
- Algunos formatos permiten la compresión, la indexación y almacenamiento y procesamiento eficientes:
 - *Avro*
 - *ORC*
 - *Parquet*:

Agenda

- Tipos de datos
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Bases de datos relacionales



Bases de datos no relacionales

- No aplican un esquema relacional se conocen como NoSQL.
- Hay 4 tipos

Bases de datos de clave-valor

Cada registro tiene una clave única y un valor asociado en cualquier formato.

Products	
Key	Value
123	"Hammer (\$2.99)"
162	"Screwdriver (\$3.49)"
201	"Wrench (\$4.25)"

Bases de datos de documentos

El valor es un documento JSON

Customers	
Key	Document
1	{ "name": "Joe Jones", "email": "joe@litware.com" }
2	{ "name": "Samir Nadoy", "email": "Samir@northwind.com" }

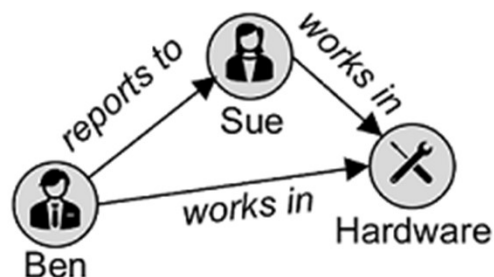
Bases de datos de familia de columnas

Ofrecen la posibilidad de dividir las columnas en grupos (familias de columnas) con una relación lógica entre sí.

Orders				
Key	Customer		Product	
	Name	Address	Name	Price
1000	Joe Jones	1 Main St.	Hammer	2.99
1001	Samir Nadoy	123 Elm Pl.	Wrench	4.25

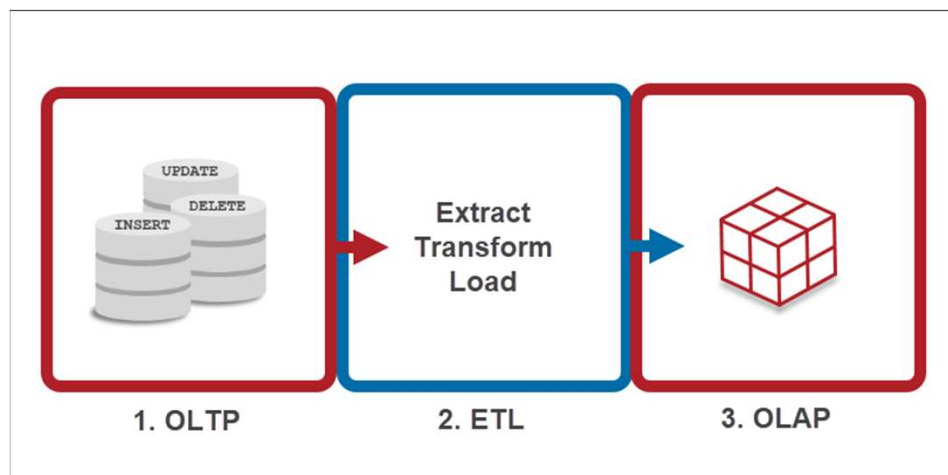
Bases de datos de grafos

Almacenan entidades como nodos con vínculos para definir relaciones entre ellas.

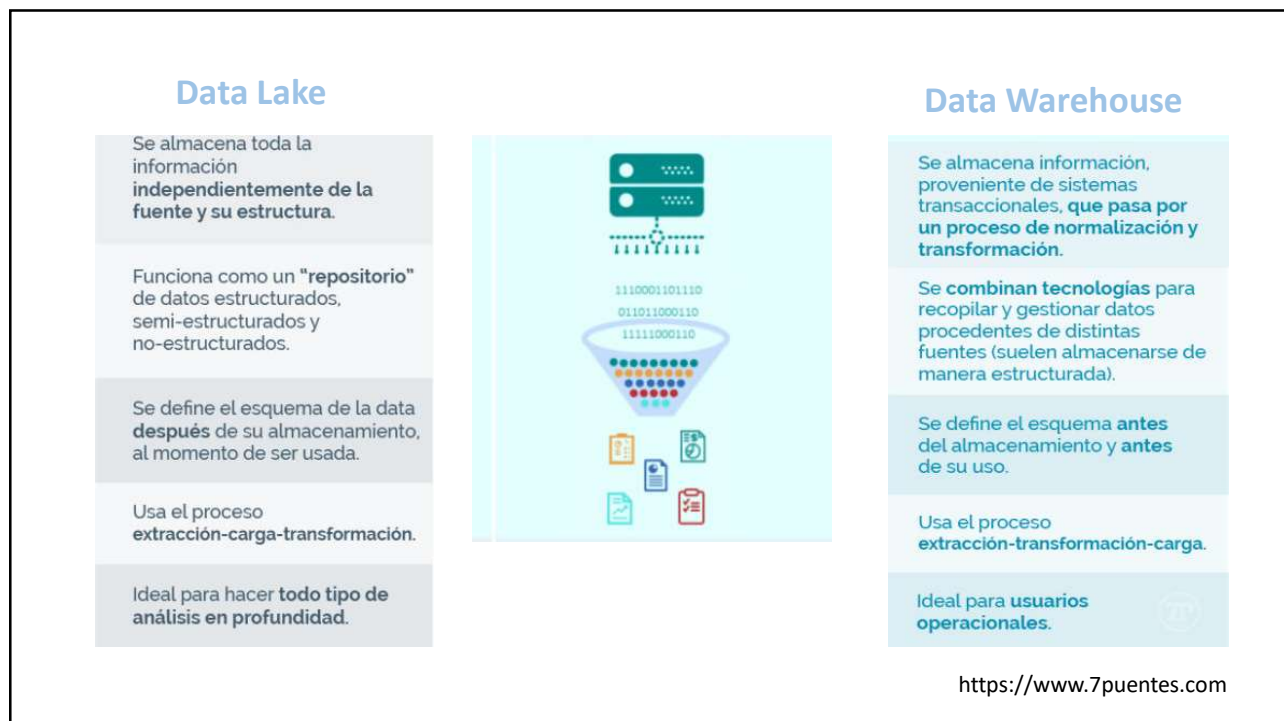
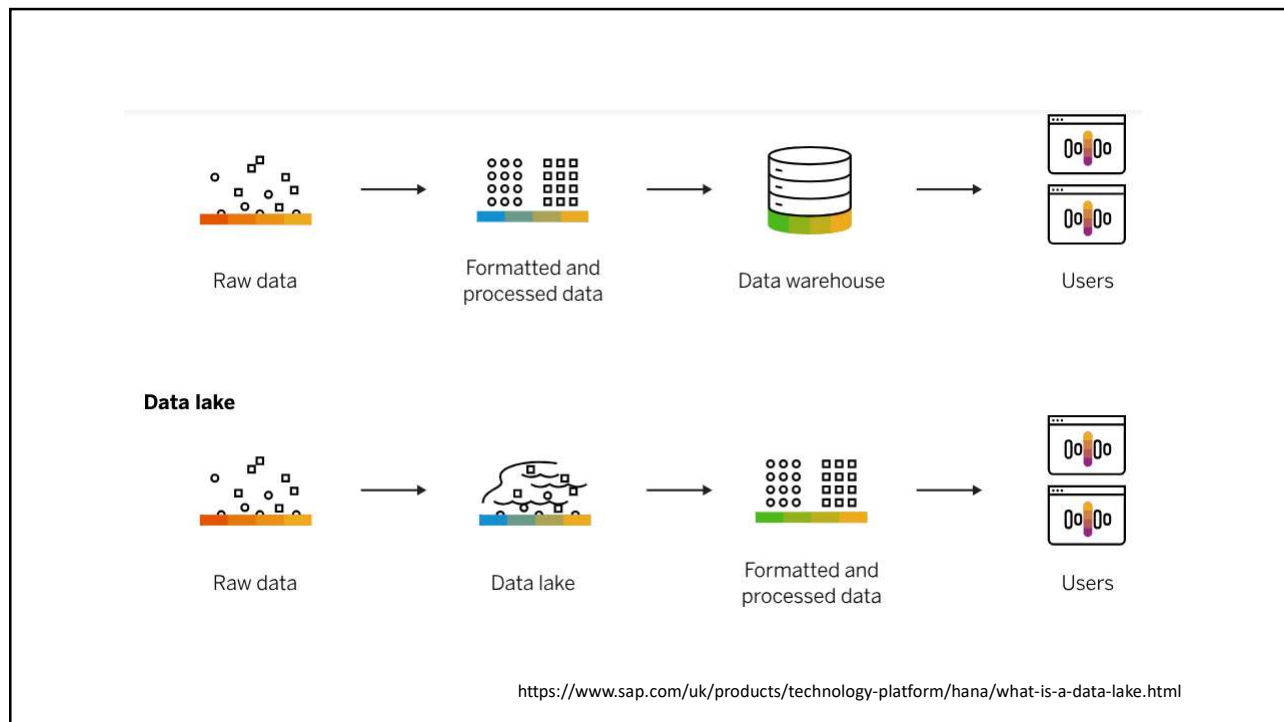


Agenda

- Tipos de datos
- Almacenes de datos
 - Almacenamiento de archivos
 - Bases de datos
 - **OLTP vs OLAP**
- Desarrollo de modelos



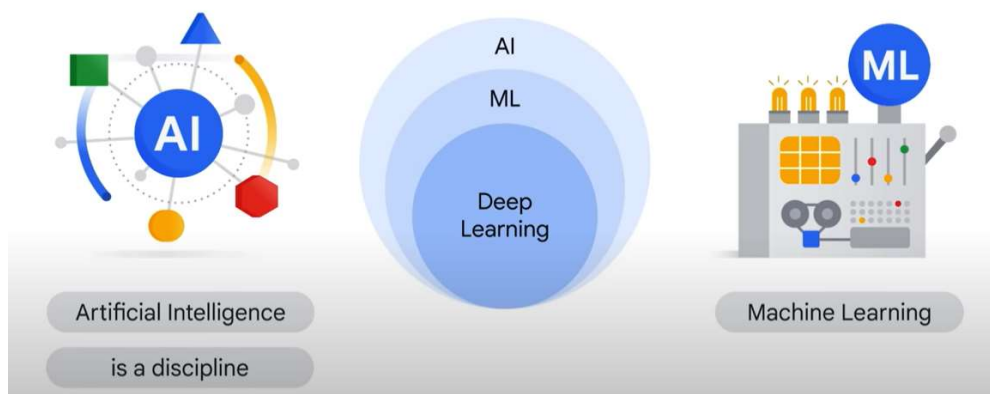
<https://phoenixnap.com>



Agenda

- Tipos de datos
- Almacenes de datos
 - Almacenamiento de archivos
 - Bases de datos
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- **Desarrollo de modelos**

Artificial intelligence



AI is the theory and development
of computer systems able to **perform tasks**
normally requiring human intelligence.

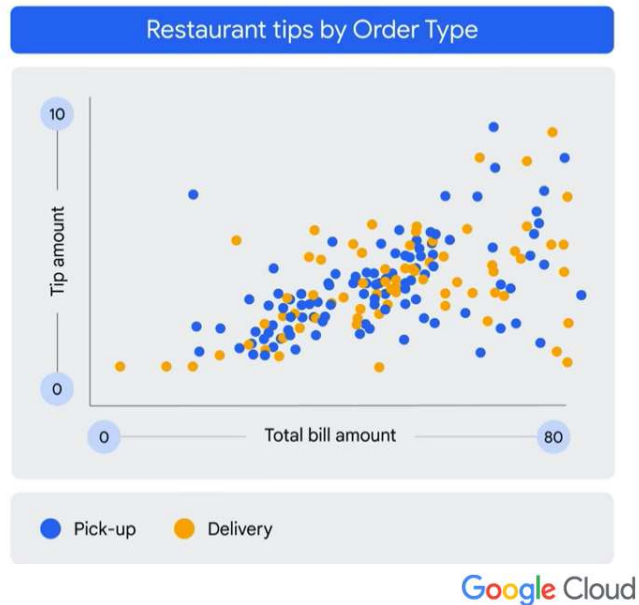
Google Cloud

ML gives computers the ability to
learn without explicit programming.

Google Cloud

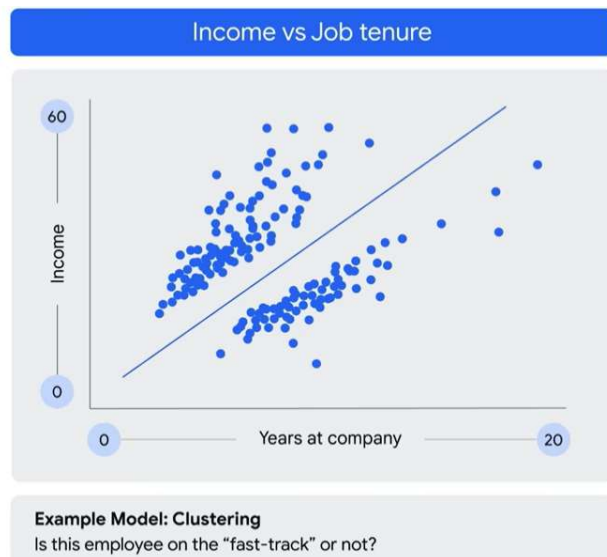
Supervised learning implies the data is already labeled

In supervised learning we are learning from past examples to predict future values.

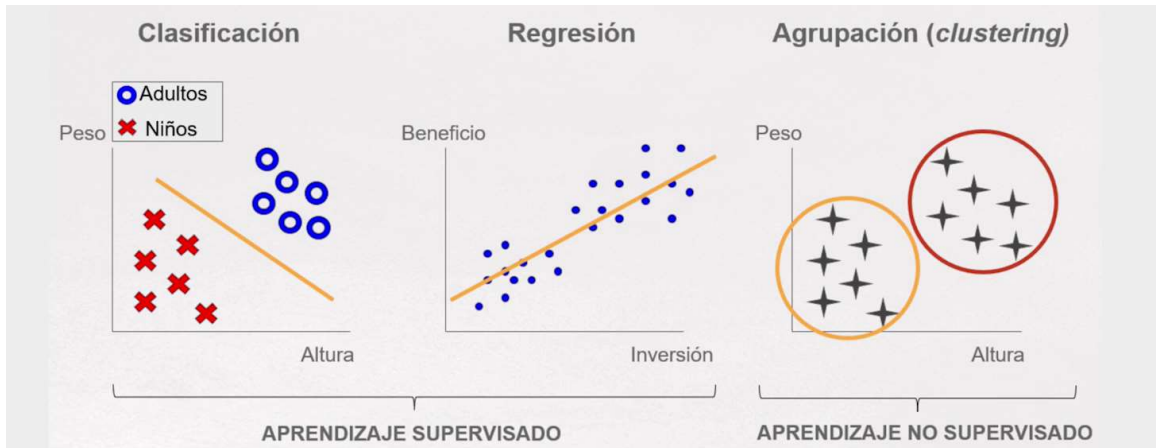


Unsupervised learning implies the data is not labeled

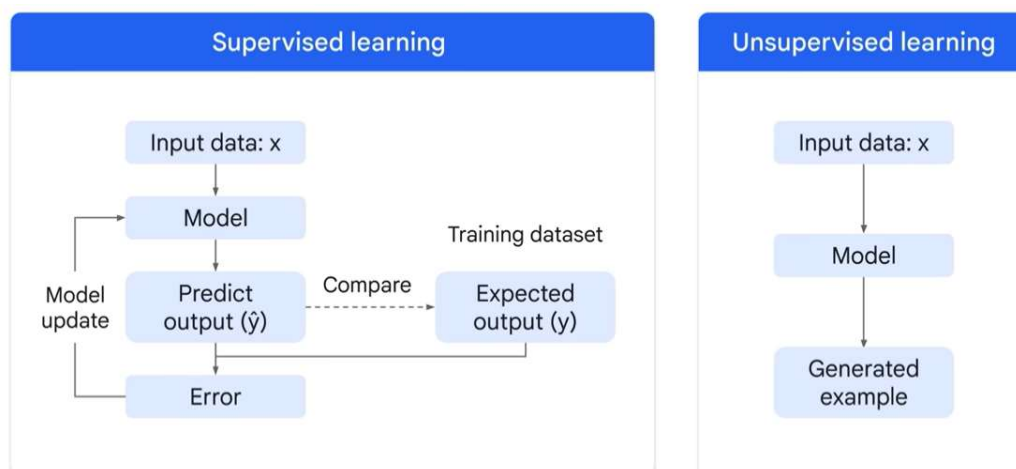
Unsupervised problems are all about looking at the raw data, and seeing if it naturally falls into groups



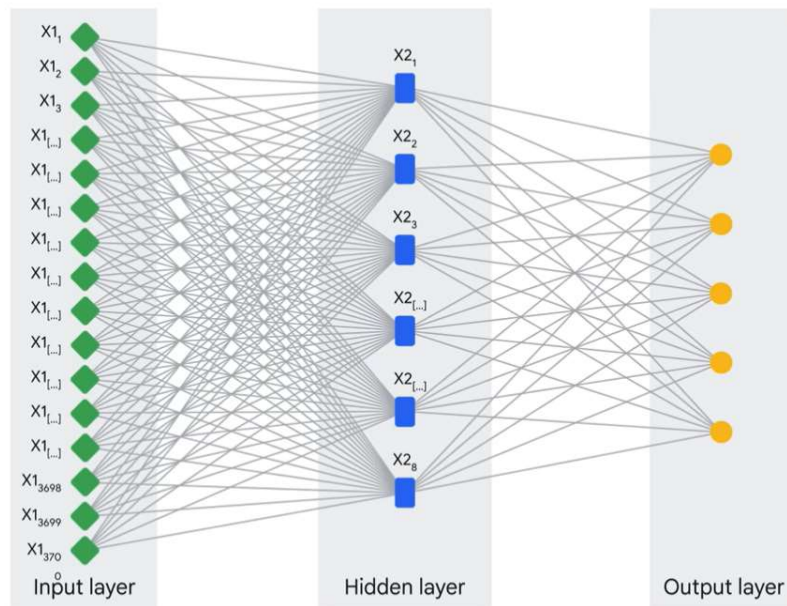
Example: PCA dimensionality reduction



<https://openwebinars.net/blog/modelos-de-machine-learning/>



Google Cloud



		Frequently used algorithms for biomedical research	Example usage (data type)
Supervised learning	Machine learning	SVM 	• Cancer vs healthy classification (gene expression)
		KNN 	• Multiclass tissue classification (gene expression)
		Regression 	• Genome-wide association analysis (SNP)
		Random forest 	• Pathway-based classification (gene expression, SNP)
	Deep learning	CNN 	• Protein secondary structure prediction (amino acid sequence)
		RNN 	• Sequence similarity prediction (nucleotide sequence)
Unsupervised learning	Clustering	Hierarchical 	• Protein family clustering (amino acid sequence)
		K-means 	• Clustering genes by chromosomes (gene expression)
	dimensionality reduction	PCA 	• Classification of outliers (gene expression)
		tSNE 	• Data visualization (single cell RNA-sequencing)
		NMF 	• Clustering gene expression profiles (gene expression)

